

Year 10 Engineering Technology - Water Control System

Lesson plans: 60 lessons across 20 weeks

Lesson Plan Overview

These lesson plans support a 20-week Stage 5 Engineering Technology Control systems unit. They assume three lessons per week and spread theory, design, construction, testing, evaluation and folio evidence across the full semester.

Assumption: three lessons per week across 20 weeks. Each week below gives a practical lesson flow that can be expanded or compressed to suit timetable interruptions.

Stage 5 Outcomes

Outcome	Official outcome wording used for this unit
EGT5-SAF-01	applies risk management and safe work practices in engineering contexts
EGT5-COM-01	communicates ideas, concepts and solutions for engineering practice
EGT5-EVL-01	investigates and evaluates engineering systems and solutions
EGT5-IVT-01	explains engineering practices and the influence of technologies used in engineering industries
EGT5-GRP-01	develops and applies technical graphics in engineering contexts
EGT5-MEA-01	applies mechanical analysis and practical testing to investigate engineering concepts
EGT5-USE-01	selects and applies materials for engineering projects
EGT5-ENV-01	analyses relationships between engineering design, production and sustainability

20-Week Lesson Sequence

Week	Focus	Lesson flow	Evidence checkpoint	Outcomes
1	Unit launch, WHS and control-system context	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check	Safety notes, first project brief response and hazard list.	EGT5-SAF-01, EGT5-

		quality and respond to feedback. Focus detail: Project brief; everyday irrigation and automation; low-voltage safety; wet and dry work zones.		IVT-01, EGT5-COM-01
2	Inputs, outputs and feedback	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Input-process-output language; open-loop and closed-loop control; moisture, rain, switch and timer inputs.	IPO diagram, vocabulary notes and feedback explanation.	EGT5-EVL-01, EGT5-COM-01, EGT5-IVT-01
3	Safe testing setup	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Workshop layout; spill control; dry electronics zone; teacher checkpoints; safe tool and component handling.	Annotated safe test setup and risk-control checklist.	EGT5-SAF-01, EGT5-COM-01
4	User need and success criteria	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Garden watering problem; user needs; water efficiency; measurable success criteria and folio setup.	User-need statement, success criteria and folio structure.	EGT5-COM-01, EGT5-ENV-01
5	Sensors and actuators	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Moisture/rain sensors, switches, solenoids, pumps, indicators, relays and controllers.	Component comparison table and annotated component sketch.	EGT5-USE-01, EGT5-MEA-01, EGT5-EVL-01
6	IPO diagrams and system vocabulary	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: System boundaries; block diagrams; feedback examples; control-system language used in context.	Block diagram and glossary entries.	EGT5-COM-01, EGT5-IVT-01
7	Algorithms and control logic	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Flowchart or	Algorithm or flowchart with teacher feedback.	EGT5-GRP-01, EGT5-COM-01

		pseudocode; watering decisions; trigger thresholds; timing and reset logic.		
8	Component research and safe test sequence	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Circuit options; pump or valve options; test order; data table design; fault-log setup.	Research notes, test sequence and draft fault log.	EGT5-EVL-01, EGT5-SAF-01
9	Concept sketching	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Reservoir, tubing, plant area, sensor position, controller location and dry electronics zone.	Two concept sketches with annotations.	EGT5-GRP-01, EGT5-COM-01
10	Final design selection	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Compare concepts; choose final layout; labelled drawing; circuit/wiring plan; irrigation layout.	Selected concept, justification and labelled final design.	EGT5-GRP-01, EGT5-EVL-01
11	Build planning and approval	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Materials/component list; tools; step-by-step build sequence; updated risk controls; teacher approval.	Build plan, component list and approval checkpoint.	EGT5-SAF-01, EGT5-COM-01
12	Testing preparation	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Testing tables; performance measures; response time; water delivered; folio evidence checklist.	Test tables, evidence checklist and prototype setup plan.	EGT5-MEA-01, EGT5-EVL-01
13	Water path construction	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Build base, reservoir, tubing route and outlet; leak testing with power disconnected.	Build log, photos and leak-test record.	EGT5-SAF-01, EGT5-USE-01, EGT5-MEA-01

14	Input/control subsystem construction	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Build and check input or switch circuit; wiring checks; test triggering without water flow.	Input test data, wiring photo or diagram and troubleshooting notes.	EGT5-SAF-01, EGT5-MEA-01
15	Output subsystem testing	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Connect pump or valve; short subsystem tests; water delivery check; teacher checkpoint.	Output test data and teacher check record.	EGT5-SAF-01, EGT5-EVL-01
16	System integration and fault-finding	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Connect subsystems; diagnose leaks, wiring faults, unreliable triggering or poor delivery.	Full-system test table and fault log.	EGT5-EVL-01, EGT5-MEA-01, EGT5-COM-01
17	Reliability testing	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Repeated test cycles; response time; water delivered; consistency and failure patterns.	Repeated-test data and reliability notes.	EGT5-EVL-01, EGT5-MEA-01
18	Controlled improvements and retesting	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Modify one variable at a time; retest; compare before and after results.	Improvement record and comparison table.	EGT5-EVL-01, EGT5-COM-01
19	Sustainability and final performance review	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Water use, material choice, cost, repairability, recycling/reuse and final performance test.	Sustainability reflection and final test evidence.	EGT5-ENV-01, EGT5-EVL-01
20	Folio completion, evaluation and reflection	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Engineering report structure; peer review; final evaluation; submission and class reflection.	Completed folio/report and project reflection.	EGT5-COM-01, EGT5-EVL-01, EGT5-IVT-01

Core Resources and Adjustments

- Low-voltage power only; teacher-approved controller/relay/circuit option; sensors or switches; pump or solenoid valve; reservoir; tubing; base board; plant-bed tray; measuring cylinder; timer; towels/spill kit and PPE.
- Student scaffolds: IPO diagram, algorithm flowchart, wiring checklist, test data table, fault log, build log and evaluation prompts.
- Adjustments: provide labelled diagrams, vocabulary banks, partially completed schematics, paired testing roles and simplified teacher-approved circuit options.
- Extension: add closed-loop feedback, compare alternative sensors/outputs, improve water efficiency or graph repeated test data.